

Sara Klaasen

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Education

PhD	ETH Zurich , Geophysics - Eidgenössische Technische Hochschule Zürich (ETH Zürich, Swiss Federal Institute of Technology Zurich) - ETH medal	Zurich, Switzerland 2020 – 2025
Msc	ETH Zurich , Geophysics	Zurich, Switzerland 2018 – 2020
Bsc	University College Roosevelt (UCR) , Liberal Arts and Sciences - Magna cum Laude - Major in Earth sciences, biology, mathematics and physics	Middelburg, Netherlands 2014 – 2017

Experience

Australian National University , Postdoctoral researcher Researching DAS projects in New Zealand. Imaging Taupo Volcano.	Canberra, Australia 2026 – present 1 year
ETH Zurich , Postdoctoral researcher	Zurich, Switzerland 2025 – 2026 1 year
Sport, Recreatie en Onderwijsvoorzieningen (SRO) , Admin Online administration	Haarlem, Netherlands 2017 – 2017 1 year
Provinciale Zeeuwse Courant (PZC) , Online editor - Online content management of the provincial newspaper - Writing and editing news articles	Vlissingen, Netherlands 2015 – 2017 2 years

Awards

ETH Medal ETH Zurich	2026
Outstanding Student Presentation Award (OSPA) Seismological Society of America Photonics Meeting	2024

Grants

Swiss National Science Foundation (SNF) postdoc.mobility grant Swiss National Science Foundation	2025
Women-in-STEM grant Fix the Leaky Pipeline program, ETH Zürich	2021
VSBFonds grant VSBFonds	2018

Research collaborations

- Australian Research Council (ARC) Centre of Excellence in Optical Microcombs for Break-through Science (COMBS)** 2026 – present
Associate investigator
www.combs.org.au
- Seismicity on Venus: Prediction and Detection** 2022 – 2024
ECR member
teams.issibern.ch/seismicityonvenus

Teaching experience

- Data Analysis and Visualisation with Python in Earth Sciences** 2022 – 2025
Teaching assistant
- Geophysical field lab** 2021 – 2023
Teaching assistant
- Dynamische Erde I** 2020 – 2023
Guest lecturer

Research Output

Book Chapters

- The leading edge of volcanology: Advances and innovations in monitoring and interpreting volcanic activity** 2027
Michael P Poland, Elske de Zeeuw-van Dalfsen, Koki Aizawa, Sara Klaasen, Fiona D'Arcy, Emma J Nicholson, Kye Anderson
(Elsevier)
- Fiber optic sensing for volcano monitoring and imaging volcanic processes** 2025
Philippe Jousset, Gilda Currenti, Shane Murphy, Eva Eibl, Corentin Caudron, Lise Retailleau, Christopher Wollin, Sara Klaasen, Takeshi Nishimura, Andreas Fichtner, Zack Spica, Arnaud Lemarchand, Charlotte Krawczyk
doi.org/10.1007/978-3-031-86841-2_17 (Springer)

Journal Articles

- Submarine seismicity monitoring with distributed acoustic sensing near Santorini and Kolumbo Volcano** 2026
Sara Klaasen, Thomas Hudson, Sanchit Sachdeva, Paraskevi Nomikou, Andreas Fichtner
doi.org/10.26443/seismica.v5i1.1845 (Seismica)
- Seismic constraints on glacier density** 2025
Ariane Lanteri, Scott Keating, Lars Gebraad, Sara Klaasen, Marta Pienkowska-Cote, Olaf Eisen, Andrea Zunino, Kristín Jónsdóttir, Coen Hofstede, Dimitri Zigone, Andreas Fichtner
doi.org/10.1038/s41598-025-26440-z (Scientific Reports)
- The Seismogenic Thickness of Venus** 2025
Julia Maia, Ana-Catalina Plesa, Iris van Zelst, Richard Ghail, Anna Gülcher, Mark Panning, Sven Peter Näsholm, Barbara de Toffoli, Anna Horleston, Krystyna Smolinski, Sara Klaasen, Robert Herrick, Raphael Garcia
doi.org/10.1038/s41598-025-26440-z (Journal of Geophysical Research: Planets)
- Towards a widely applicable earthquake detection algorithm for fibreoptic and hybrid fibre-optic-seismometer networks** 2025
Thomas Hudson, Sara Klaasen, Olivier Fontaine, Conor Bacon, Kristin Jonsdottir, Andreas Fichtner
doi.org/10.1093/gji/ggae459 (Geophysical Journal International)

- An illustrated guide to: Distributed and integrated fibre-optic sensing in seismology** 2024
Andreas Fichtner, Fabian Walter, Patrick Paitz, Sara Klaasen, Daniel Bowden, Sebastian Noe, Nils Müller, Dominik Husmann, Jacques Morel
[dx.doi.org/10.1016/j.jeqs.2024.09.006](https://doi.org/10.1016/j.jeqs.2024.09.006) (Earthquake Sciences)
- Seismic wave detectability on Venus using ground deformation sensors, infrasound sensors on balloons and airglow imagers** 2024
Raphael Garcia, Iris van Zelst, Taichi Kawamura, Sven Peter Näsholm, Anna Horleston, Sara Klaasen, Maxence Lefevre, Celine Marie Solberg, Krystyna Smolinski, Ana-Catalina Plesa, Quentin Brissaud, Julia Maia, Simon Stähler, Philippe Lognonné, Mark Panning, Anna Gülcher, Richard Ghail, Barbara de Toffoli
doi.org/10.1029/2024EA003670 (Earth and Space Science)
- Modelling uncertainty in P-wave arrival-times retrieved from DAS data: case-studies from 15 fibre optic cables** 2024
Emanuele Bozzi, Nicola Piana Agostinetti, Andreas Fichtner, Sara Klaasen, Arantza Ugalde, Biondo Biondi, S Yuan, Torsten Dahm, Marius Isken, Patrick Paitz, Fabian Walter, Alain Baird, Carlos Becerril, Takeshi Nishimura, Junzhu Shen, Tieyuan Zhu, Gilberto Saccorotti
doi.org/10.1093/gji/ggae364 (Geophysical Journal International)
- Complex spatial distribution of onset amplitude and waveform correlation: case studies from different DAS experiments** 2024
Emanuele Bozzi, Gilberto Saccorotti, Nicola Piana Agostinetti, Carlos Becerril, Andreas Fichtner, Sara Klaasen, Takeshi Nishimura, Junzhu Shen, Fabian Walter, Tieyuan Zhu
doi.org/10.4430/bgo00458 (Bulletin of geophysics and oceanography)
- Challenges in submarine fiber-optic earthquake monitoring** 2024
Jonas Igel, Sara Klaasen, Sebastian Noe, Paraskevi Nomikou, Konstantinos Korantzalos, Andreas Fichtner
doi.org/10.1029/2024JB029556 (Journal of Geophysical Research: Solid Earth)
- Subglacial volcano monitoring with fibre-optic sensing: Grímsvötn, Iceland** 2023
Sara Klaasen, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristín Jónsdóttir, Lars Gebraad, Patrick Paitz, Andreas Fichtner
doi.org/10.30909/vol.06.02.301311 (Volcanica)
- Fiber-optic observation of volcanic tremor through floating ice sheet resonance** 2022
Andreas Fichtner, Sara Klaasen, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Patrick Paitz, Kristin Jonsdottir
doi.org/10.1785/0320220010 (The Seismic Record)
- Distributed acoustic sensing in volcano-glacial environments — Mount Meager, British Columbia** 2021
Sara Klaasen, Patrick Paitz, Nadja Lindner, Jan Dettmer, Andreas Fichtner
doi.org/10.1029/2021JB022358 (Journal of Geophysical Research: Solid Earth)
- Detecting Seismic Events with Computer Vision: Applications for Fiber-Optic Sensing** 2021
Solvi Thrastarson, Robert Torfason, Sara Klaasen, Patrick Paitz, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Andreas Fichtner
doi.org/10.1002/essoar.10509693.1 (ESS Open Archive)
- First-Author Conference Proceedings**
- Modelling months of deformation during the Sundhnúkur eruptions on the Reykjanes Peninsula, Iceland** 2026
Sara Klaasen, Daniel Bowden, Kristin Jonsdottir, Yesim Cubuk-Sabuncu, Jack Maxwell, Meghan Miller, Andreas Fichtner
(SSA optical seismology and the next era of seismic sensing)

Submarine volcano monitoring with Distributed Acoustic Sensing at Kolumbo, Greece Sara Klaasen, Thomas Hudson, Paraskevi Nomikou, Andreas Fichtner (IAVCEI General Meeting)	2025
Full-Waveform Inversion of DAS data on a subglacial volcano: Grímsvötn, Iceland Sara Klaasen, Sebastian Noe, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Andreas Fichtner (IAVCEI General Meeting)	2025
Submarine volcano monitoring with Distributed Acoustic Sensing at Kolumbo, Greece Sara Klaasen, Thomas Hudson, Paraskevi Nomikou, Andreas Fichtner (SSA General Meeting)	2025
Moment Tensor inversion with Full-Waveform Inversion and Distributed Acoustic Sensing on a subglacial volcano: Grímsvötn, Iceland Sara Klaasen, Sebastian Noe, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Lars Gebraad, Andreas Fichtner (EGU Copernicus Meetings)	2024
Submarine volcano monitoring with fiber-optic cables at Santorini Sara Klaasen, Paraskevi Nomikou, Konstantinos Karantzas, Jonas Igel, Andreas Fichtner (AGU Annual Meeting)	2023
Going towards Full Waveform Inversion (FWI) with Distributed Acoustic Sensing (DAS) on a subglacial volcano: Grímsvötn, Iceland Sara Klaasen, Sebastian Noe, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Jonas Igel, Andreas Fichtner (AGU Annual Meeting)	2023
Volcano monitoring and modelling using Distributed Acoustic Sensing (DAS) on a subglacial volcano: Grímsvötn, Iceland Sara Klaasen, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Lars Gebraad, Sebastian Noe, Andreas Fichtner (IUGG General Assembly)	2023
Monitoring glacially covered volcanoes with Fibre-Optic Sensing Sara Klaasen, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Lars Gebraad, Andreas Fichtner (IAVCEI General Meeting)	2023
Volcanic Event Characterization with Fiber-Optic Sensing Sara Klaasen, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Lars Gebraad, Andreas Fichtner (AGU Annual Meeting)	2022
Fibre-Optic Sensing for Volcano Monitoring on Grímsvötn, Iceland Sara Klaasen, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Lars Gebraad, Andreas Fichtner (EGU General Assembly)	2022
Distributed Acoustic Sensing on Volcanoes: Grímsvötn and Fagradalsfjall, Iceland Sara Klaasen, Solvi Thrastarson, Kristin Jonsdottir, Yesim Cubuk-Sabuncu, Patrick Paitz, Andreas Fichtner (AGU Annual Meeting)	2021

Combining Distributed Acoustic Sensing and Beamforming in a Volcanic Environment on Mount Meager, British Columbia Sara Klaasen, Patrick Paitz, Jan Dettmer, Andreas Fichtner (SSA General Meeting)	2021
Combining Distributed Acoustic Sensing and Beamforming in a Volcanic Environment on Mount Meager, British Columbia Sara Klaasen, Patrick Paitz, Jan Dettmer, Andreas Fichtner (EGU General Assembly)	2021
Distributed Acoustic Sensing for the Exploration of the Mount Meager Volcanic Complex, British Columbia, Canada Sara Klaasen, Andreas Fichtner, Patrick Paitz, Jan Dettmer (AGU Annual Meeting)	2020

Invited Talks

Keck observatory workshop SSA optical seismology and the next era of seismic sensing, Kona (Pre-conference workshop)	2026
Frontiers in Fiber-Optics Sensing in Seismology IASPEI-IAGA, Lisbon (Conference session)	2025
ECR Keynote talk IAVCEI, Geneva (Conference keynote)	2025
Seminar Stanford University, San Francisco (Seminar)	2025
Seminar University of Washington, online (Seminar)	2024
Session: An innovative photonic vision of volcanoes and geothermal systems SSA Photonic Seismology, Vancouver (Session)	2024
Colloquium CSIRO, Perth (Seminar)	2023
Geophysics Weekly Meeting Australian National University, Canberra (Seminar)	2023
Seminar University of Canterbury, Christchurch (Seminar)	2023
Workshop on Distributed Acoustic Sensing AGU, online (Pre-conference workshop)	2021

Service

Convening conference sessions

IAVCEI, AGU

Peer-review

Seismica, SRL

Languages

Dutch

native

English

C2

German

B2

French

B1